1. Elaborate PDDL.

PDDL (Planning Domain Definition Language) is a formal language used to describe planning problems in Artificial Intelligence. It specifies the objects, actions, initial state, and goal state of a problem.

PDDL consists of two main descriptions:

- Domain description: Defines predicates, action schemas, preconditions, and effects.
- Problem description: Defines the objects, initial state, and goal conditions.

Example: In a tire-changing problem, Remove(Flat, Axle) is an action that removes the flat tire from the axle.

An action is generally represented using:

- Preconditions — conditions required to execute the action.
- Effects — changes produced by the action.

PDDL allows an AI planner to search for a sequence of actions that transforms the initial state into the goal state.

2. Explain Classical Planning with the Spare Tire Problem.

Classical planning is a planning approach in which an agent finds a sequence of actions to transform a known initial state into a desired goal state. It assumes a fully observable, deterministic, and static environment during planning, with discrete states and actions.

Spare tire problem:

- Initial state: A flat tire is mounted on the axle, and a good spare tire is in the trunk.
- Goal state: The good spare tire is mounted on the axle.
- Actions: Remove(Flat, Axle), Remove(Spare, Trunk), and PutOn(Spare, Axle).

Solution plan:

1. Remove the flat tire from the axle.
2. Remove the spare tire from the trunk.
3. Mount the spare tire on the axle.

The action LeaveCarOvernight is avoided because it causes the tires to disappear. This example demonstrates how planning selects actions to achieve a goal while avoiding undesirable effects.

3. Describe Planning Graphs for Heuristic Estimation.

A planning graph is a layered structure that represents the possible states and actions reachable from an initial state. It is used by the GRAPHPLAN algorithm and helps estimate the cost of achieving goals.

Planning graphs contain:

- State levels (S_i): Represent possible literals or facts.
- Action levels (A_i): Represent actions applicable at the preceding state level.
- Mutex relations: Identify actions or literals that cannot occur together.

Heuristic estimation:

1. Construct the planning graph from the initial state.
2. Find the first level at which each goal literal appears.
3. Use this level as the literal's level-cost heuristic.

If a goal literal never appears even after the graph levels off, the problem is unsolvable. The graph helps estimate goal-achievement costs and guides efficient search.

4. How do you represent a set of ground actions?

A ground action is an action in which all variables have been replaced by specific objects or constants.

A set of ground actions is obtained by instantiating each action schema with all valid combinations of objects in the domain. This process is called propositionalization.

Example: Consider the action schema:

Fly(p, from, to)

Possible ground instances are:

- Fly(P1, SFO, JFK)
- Fly(P2, JFK, SFO)

Only instances whose preconditions are satisfied are applicable in the current state.

The complete set of applicable actions in state s is represented as:

$$ACTIONS(s) = \{a \mid s \models PRECOND(a)\}$$

Ground actions allow a planning problem to be represented as a propositional state-space search problem.

5. Define Schema.

An action schema is a generalized representation of an action containing variables, preconditions, and effects. It represents a family of ground actions.

Example:

Fly(p, from, to)

- Preconditions: The object p is a plane located at from, and both locations are airports.
- Effects: The plane is at to and is no longer at from.

By substituting specific values for the variables, the schema generates ground actions such as Fly(P1, SFO, JFK).

Action schemas reduce the need to specify every possible action separately.

6. Define Levels in Planning Graphs.

A planning graph consists of alternating state and action levels.

- State level (S_i): Contains literals that may hold at that stage.
- Action level (A_i): Contains actions whose preconditions are satisfied at (S_i).
- Next state level (S_{i+1}): Contains the possible effects of actions in A_i, including facts that persist.

The graph starts with S_0, representing the initial state, and expands as $S_0, A_0, S_1, A_1, \ldots$.

Mutex links indicate mutually exclusive actions or literals. Expansion continues until the graph levels off or a solution is found.

7. Give the differences between Forward and Backward State-Space Search.

Forward State-Space Search	**Backward State-Space Search**
Starts from the initial state.	Starts from the goal state or goal conditions.
Applies actions to generate successor states.	Identifies actions that can achieve the current goal.
Also called progression planning.	Also called regression planning.
Searches toward the goal.	Searches toward the initial state.

Example: In the spare tire problem, forward search starts with the flat tire on the axle, whereas backward search starts with the goal of mounting the good spare tire and identifies the actions needed to achieve it.

8. Define Optimistic Description.

An optimistic description is an approximate representation of a high-level action that includes all possible favorable effects, even if those effects cannot occur together in a single execution.

It is used in hierarchical planning to estimate reachable states without exploring every detailed implementation.

Example: A high-level action Go(Home, SFO) may optimistically be described as both reaching the destination and retaining cash, even though a particular implementation may require spending cash.

Optimistic descriptions help planners quickly determine whether a high-level plan might achieve the goal. However, further refinement is required to verify that an executable plan actually works.

9. Describe Planning Approaches.

Planning approaches are methods used by AI systems to find a sequence of actions that achieves a goal.

Important approaches include:

1. State-space planning: Searches through states using forward progression or backward regression.
2. Planning-graph approach: Uses state levels, action levels, and mutex relations to identify possible plans.
3. Hierarchical planning: Decomposes high-level tasks into smaller, executable actions.
4. SAT-based planning: Converts a planning problem into a Boolean satisfiability problem.

These approaches aim to reduce the search space and find efficient plans while satisfying the goal conditions.

10. Define Planning.

Planning is the process of finding a sequence of actions that transforms an initial state into a desired goal state.

A planning problem consists of:

- Initial state: The current situation of the environment.
- Actions: Operations with preconditions and effects.
- Goal state: The conditions that must be achieved.

Example: In the spare tire problem, planning identifies the actions needed to remove the flat tire and mount the spare tire.

The resulting sequence of actions is called a plan.

11. Define Preconditions and Effects.

Precondition: A condition that must be satisfied before an action can be executed.

Effect: The change in the state of the environment resulting from an action.

Example: Consider the action Fly(P1, SFO, JFK).

- Precondition: At(P1, SFO) — the plane must be at SFO.
- Effects: At(P1, JFK) becomes true, and At(P1, SFO) becomes false.

An action can be executed only when its preconditions hold, and its effects determine the resulting state.

12. Define Coordination.

Coordination is the process of organizing the activities of multiple agents so that they work together effectively and avoid conflicts while achieving their goals.

It involves sharing information, allocating tasks, scheduling actions, and resolving conflicts.

Example: In a team of delivery robots, coordination ensures that robots deliver packages efficiently without colliding or unnecessarily visiting the same locations.

Coordination is essential in multi-agent systems where agents' actions may affect one another.

13. Define Multi-Agent Planning.

Multi-agent planning is the process of developing plans for multiple agents operating in the same environment. Agents may cooperate, compete, or pursue individual goals.

Important aspects include:

- Task allocation among agents.
- Coordination of actions and resources.
- Avoiding conflicts between agents.
- Achieving individual or shared goals.

Example: Multiple warehouse robots coordinate their movements and tasks to transport goods efficiently.

Multi-agent planning can be centralized, where one planner controls all agents, or decentralized, where agents develop their own plans and communicate as needed.

14. Define Hierarchical Planning.

Hierarchical planning is an AI planning approach that solves complex problems by decomposing high-level tasks into smaller subtasks until they become executable actions.

It is commonly implemented using Hierarchical Task Network (HTN) planning.

Example: The high-level task GoOnVacation may be decomposed into:

1. Book transportation.
2. Reserve accommodation.
3. Travel to the destination.

Each subtask can be further decomposed into more detailed actions.

Hierarchical planning reduces complexity by reasoning at multiple levels of abstraction.

15. Define High-Level Actions.

High-Level Actions (HLAs) are abstract actions that represent complex tasks and can be refined into sequences of lower-level actions.

They are used in hierarchical planning to simplify problem-solving.

Example: Go(Home, SFO) is a high-level action that may be implemented using actions such as:

- Travel to the airport.
- Check in.
- Board a flight.
- Fly to SFO.

An HLA can have multiple possible implementations. Refining an HLA into smaller tasks is called task decomposition.

16. Define Progression and Regression.

Progression: The process of moving forward from the initial state by applying applicable actions to generate successor states.

$$Result(s, a) = (s - DEL(a)) \cup ADD(a)$$

Here, $DEL(a)$ is the set of facts deleted and $ADD(a)$ is the set of facts added by action a.

Regression: The process of working backward from the goal by identifying actions that can achieve the goal and determining the conditions that must have held before those actions.

Progression	**Regression**
Starts from the initial state.	Starts from the goal conditions.
Applies actions forward.	Reasons backward through actions.
Generates successor states.	Generates predecessor goal conditions.

17. Define Decomposition.

Decomposition is the process of breaking a complex, high-level task into smaller subtasks that are easier to plan and execute.

It is a fundamental concept in hierarchical planning.

Example: The task Organize a Conference can be decomposed into:

1. Book the venue.
2. Invite speakers.
3. Register participants.
4. Arrange equipment.

Each subtask may be further divided into simpler tasks until all required actions are executable.

Decomposition reduces planning complexity and improves the organization of complex tasks.

18. What is State-Space Search?

State-space search is a planning technique in which each node represents a state of the environment and each edge represents an action that transforms one state into another.

The planner searches for a path from the initial state to a state satisfying the goal conditions.

Two major methods are:

- Forward search: Expands states from the initial state.

- Backward search: Works backward from the goal conditions.

Example: In the spare tire problem, each state describes the locations of the flat and spare tires, and actions connect the initial state to the goal state.

19. Define a Planning Graph.

A planning graph is a layered graph consisting of alternating proposition (state) levels and action levels. It represents which facts and actions may be possible as planning progresses.

Its main components are:

- Proposition levels: Contain literals that may hold.
- Action levels: Contain applicable actions and persistence actions.
- Mutex links: Represent pairs of actions or literals that cannot occur together.

The graph is used by the GRAPHPLAN algorithm to search for plans and estimate the cost of achieving goals.

20. What Is the Main Goal of Regression Planning?

The main goal of regression planning is to find a sequence of actions that achieves the desired goal by reasoning backward toward the initial state.

It works as follows:

1. Start with the goal conditions.
2. Select an action whose effects achieve some required goal conditions without contradicting them.
3. Replace the achieved conditions with the action's preconditions to obtain the regressed goal.
4. Continue until the regressed conditions are satisfied by the initial state.

Example: To achieve SpareOnAxle, regression identifies PutOn(Spare, Axle) and works backward to determine that the spare tire must be available and the axle must be free.

Thus, regression planning searches backward from the goal to identify a valid plan.